

EX NO:	Design of Low Pass Filter using Windowing Techniques
DATE:	4.1.Design and Analysis of an FIR Low-Pass Filter Using the Rectangular Window Technique in MATLAB.

PROGRAM

```

clc; clear; close all;

%% Input Parameters
fp = 1000; % Passband cutoff frequency (Hz) fs = 4000; % Sampling
frequency (Hz)
N = 50; % Filter order

wc = fp/(fs/2); % Normalized cutoff frequency

%% FIR Low Pass using Rectangular Window b_rect =
fir1(N,wc,'low',rectwin(N+1));

disp('--- Rectangular Window Coefficients ---'); disp(b_rect');

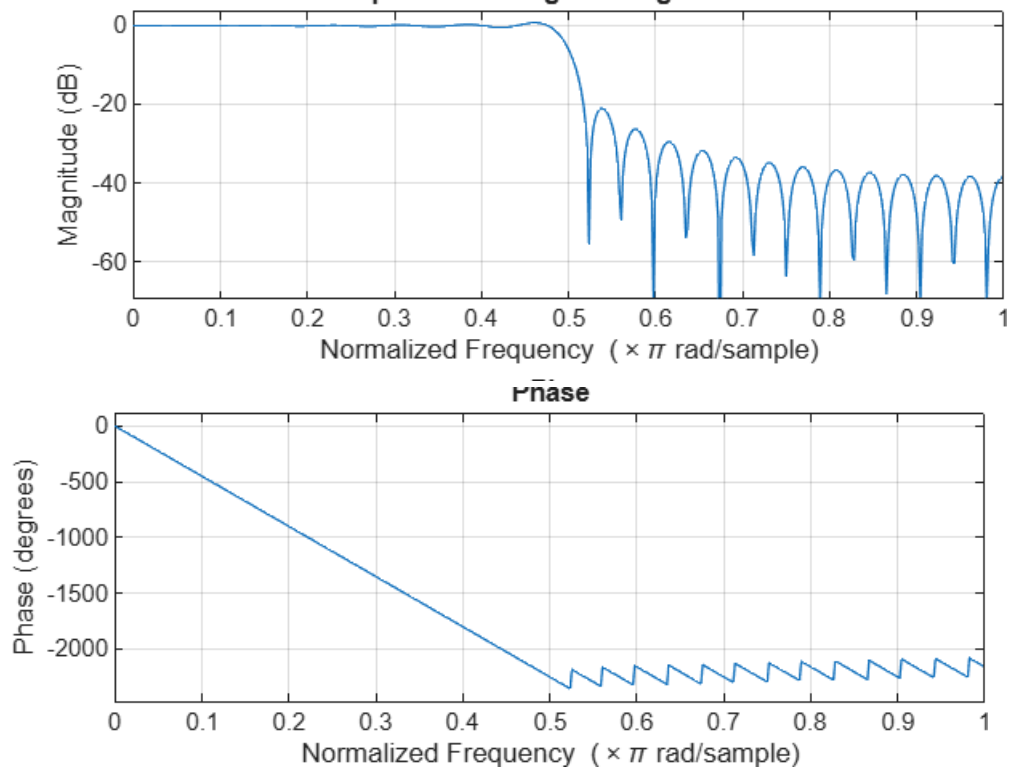
%% Frequency Response figure;
freqz(b_rect,1);
title('Low-pass FIR using Rectangular Window');

```

--- Rectangular Window Coefficients ---

```
0.0126
0
-0.0137
0
0.0150
0
-0.0166
0
0.0185
0
-0.0210
0
0.0242
0
-0.0286
0
0.0349
0
```

Low-pass FIR using Rectangular Window



Ex no	Design of Low Pass Filter using Windowing Techniques
date	4.2. Design and Analysis of an FIR Low-Pass Filter Using the Hamming Window Technique

PROGRAM

```

clear;
close all;

%% Input Parameters
fp = 1000; % Passband cutoff frequency (Hz) fs = 4000; % Sampling
frequency (Hz)
N = 50; % Filter order

wc = fp/(fs/2); % Normalized cutoff frequency

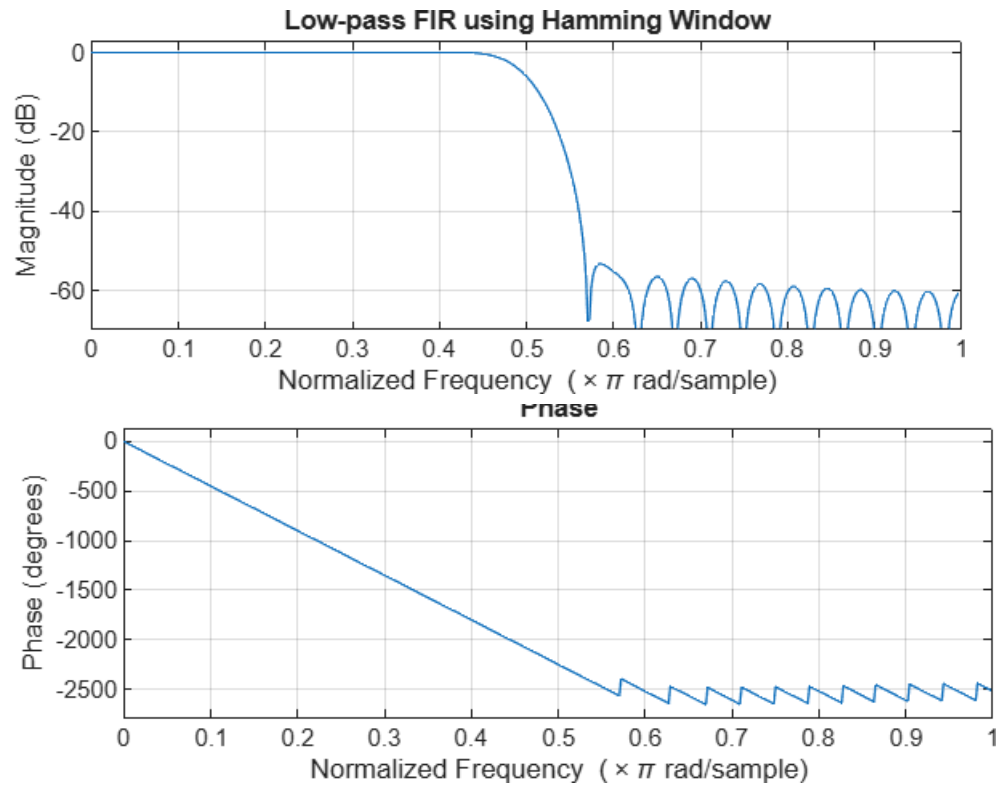
%% FIR Low Pass using Hamming Window b_hamm =
fir1(N,wc,'low',hamming(N+1));

disp('--- Hamming Window Coefficients ---'); disp(b_hamm);

%% Frequency Response figure;
freqz(b_hamm,1);
title('Low-pass FIR using Hamming Window');

--- Hamming Window Coefficients ---
0
-0.0084
0
0.0125
0
-0.0181
0
0.0260
0
-0.0379
0
0.0580
0
-0.1026
0
0.3168
0.4995
0.3168

```



--- Hamming Window Coefficients ---

```

0.0010
0
-0.0013
0
0.0021
0
-0.0034
0
0.0055
0
-0.0084
0
0.0125
0
-0.0181
0
0.0260
0

```

Ex no	Design of Low Pass Filter using Windowing Techniques
Date	4.3. Design and Analysis of an FIR Low-Pass Filter Using the Hanning Window Technique

PROGRAM

```
clc; clear; close all;

%% Input Parameters
fp = 1000; % Passband cutoff frequency (Hz) fs = 4000; % Sampling
frequency (Hz)
N = 50; % Filter order

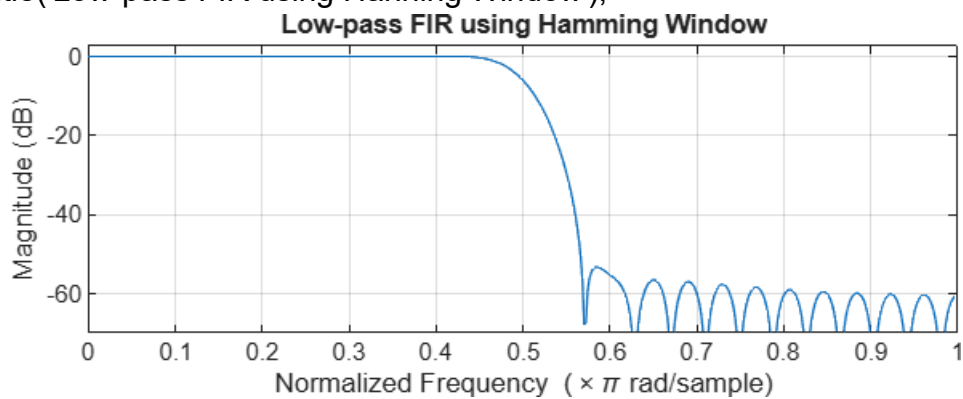
wc = fp/(fs/2); % Normalized cutoff frequency

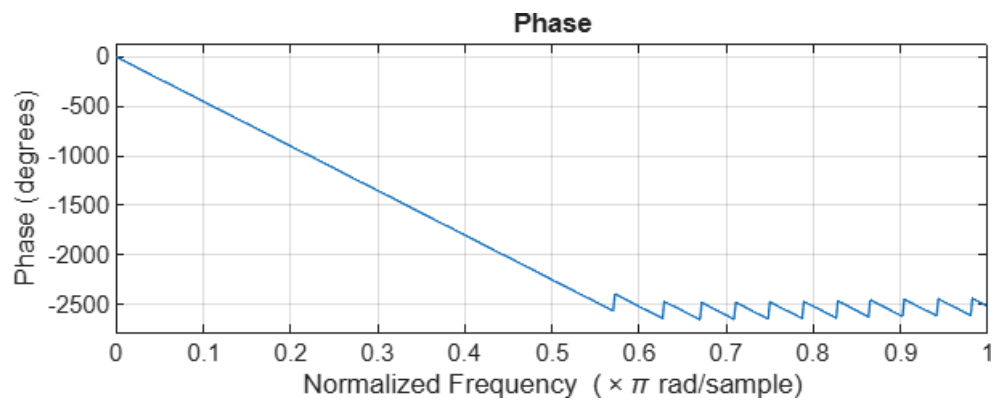
%% FIR Low Pass using Hanning Window b_hann =
fir1(N,wc,'low',hanning(N+1));

disp('--- Hanning Window Coefficients ---'); disp(b_hann');

%% Frequency Response figure;
freqz(b_hann,1);

title('Low-pass FIR using Hanning Window');
```





--- Hamming Window Coefficients ---

0.0010	-0.0181
0	0
-0.0013	0.0260
0	0
0.0021	-0.0379
0	0
-0.0034	0.0580
0	0
0.0055	-0.1026
0	0
-0.0084	0.3168
0	0.4995
0.0125	0.3168
0	0
-0.0181	-0.1026
0	0
0.0260	0.0580
0	0

--- Hamming Window Coeffic

0
0.0260
0
-0.0181
0
0.0125
0
-0.0084
0
0.0055
0
-0.0034
0
0.0021
0
-0.0013
0
0.0010